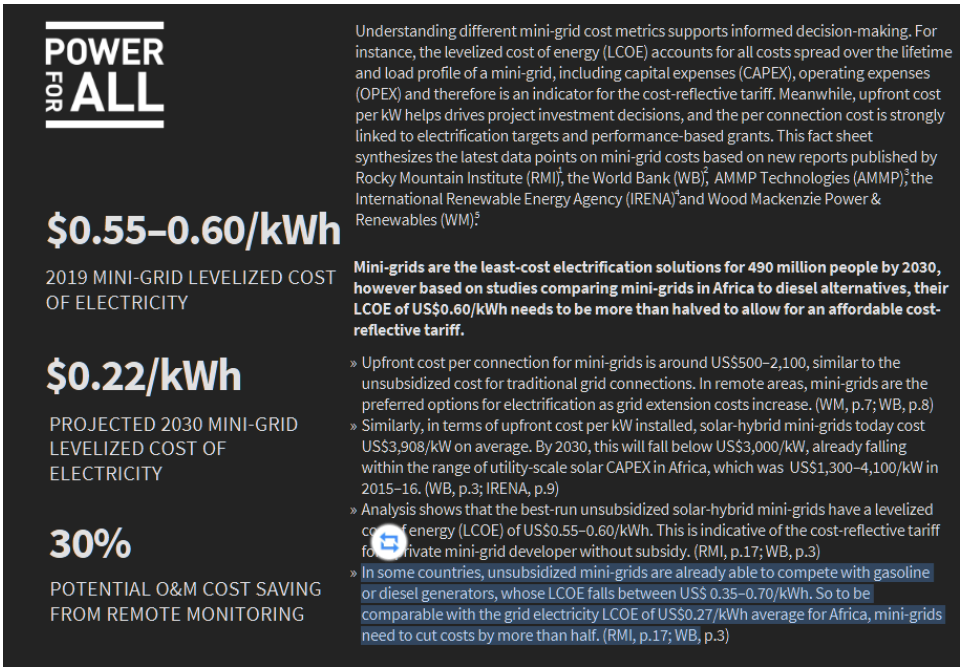
Reference 06: Baseline Diesel Electricity Cost & Fuel Crisis

1. What is this? (Component & Checked Cost)

- **Component:** Displaced diesel generation electricity cost (fuel, maintenance, generator replacement)
- **Checked Cost:** $\$0.50\ \text{USD/kWh}$ (Standard off-grid mini-grid range: $\$0.35–\$0.70/\text{kWh}$)
- **Baseline Annual Expenditure:** $\$2,920\ \text{USD/year}$ ($5,840\ \text{kWh/yr} \times \$0.50$)

2. The Proof (Screenshots & Economic Data)

Evidence Source	Screenshot Proof
World Bank / ESMAP Diesel Benchmark	

Evidence Source	Screenshot Proof
UNDP Sudan Fuel Crisis Report	

> **World Bank / ESMAP / Power for All Finding:** Standalone diesel generation in off-grid African rural communities has an unsubsidized Levelized Cost of Electricity (LCOE) of $\$0.35–\$0.70/\text{kWh}$, heavily driven by fuel delivery and low generator loading efficiency.

> **UNDP Sudan Market Report Finding:** "*Prolonged conflict has severely disrupted fuel supply chains... Diesel pump prices in remote northern and western regions have surged over 300%–500% above official rates, frequently exceeding $2.00–$3.50/L on informal supply corridors.*"

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3. How was it calculated?

- **Clinic Annual Electricity Demand:**

$$\text{Annual Energy} = 16\ \text{kWh/day} \times 365\ \text{days} = 5,840\ \text{kWh/year}$$

- **Baseline Operational Cost at $\$0.50/\text{kWh}$:**

$$\text{Baseline Cost} = 5,840\ \text{kWh} \times \$0.50/\text{kWh} = \mathbf{\$2,920}\ \text{USD/year}$$

- **Conservative Margin:**
  - Choosing $\$0.50/\text{kWh}$ sits squarely in the middle of the ESMAP benchmark, remaining realistic without relying on crisis black-market fuel spikes ($> \$1.00/\text{kWh}$).
- **Comparison to Generative AI Guess:**
  - AI guessed a flat $\$0.35/\text{kWh}$ (lower bound, assumed pre-crisis fuel delivery).
  - Grounded to $\$0.50/\text{kWh}$ based on World Bank rural minigrid data and UNDP fuel supply reality.

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4. Source Link

- **Power for All / World Bank ESMAP Fact Sheet:** Decentralized Renewable Energy Factsheet
- **UNDP Sudan Energy Assessment:** Solar Energy Value Chains in Sudan (UNDP May 2024)